



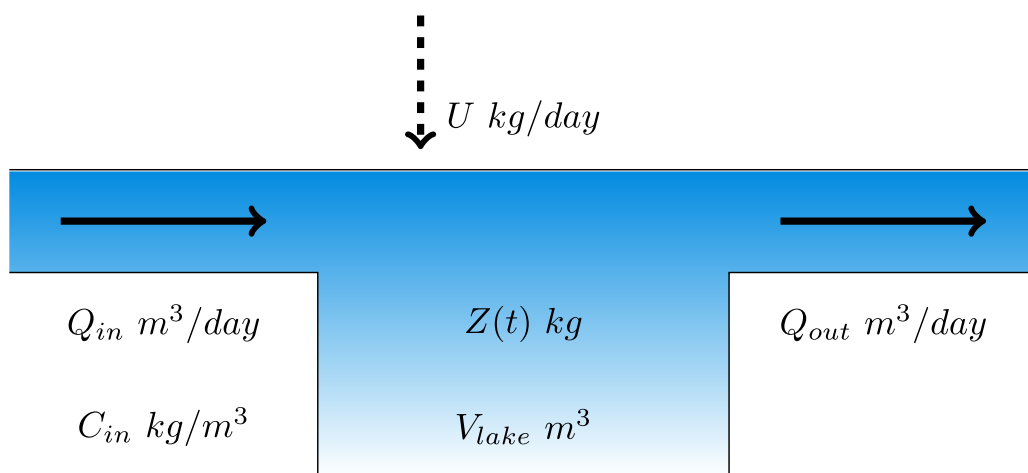
[Outline](#) > 1. Introducing mathematical modelling > 1.7 Practice problem (Mixing problem) > Validation

Validation

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In this section, we first summarise the results up till now for you, then you answer the question of the factory, and finally, you will predict what happens when the situation is reversed.

For your convenience: a summary of the results up till now:



A factory wants to start discharging waste material in a lake. The river that feeds into the lake is already polluted somewhat. How much would this new contamination increase the pollution in the lake?

A mathematical model for $Z(t)$, the mass of the pollutant in the lake, when the factory starts discharging a pollutant at $t = 0$:

$$\frac{dZ}{dt} = -\frac{Q}{V_{lake}}Z(t) + C_{in}Q + U, \quad Z(0) = C_{in}V_{lake}.$$

The solution $Z(t)$ will increase to $Z_e = \lim_{t \rightarrow \infty} Z(t) = V_{lake}(C_{in} + \frac{U}{Q})$.

To quantify the pollution increase caused by the factory, we define:

C_e : The concentration of the pollution in the lake for $t \rightarrow \infty$ when the factory discharges U kilo of pollutant per day.

C_0 : The final concentration of the pollution in the lake for $t \rightarrow \infty$ when the factory does not pollute the lake.

The waste of the factory increases the pollution concentration in the lake with the factor: $\frac{C_e}{C_0}$.

Exercise A

1/1 point (graded)

An expression for $\frac{C_e}{C_0}$ is:

☐ $1 + \frac{U}{Q}$

☐ $1 + \frac{U}{C_{in} V_{lake}}$

☐ $1 + \frac{V_{lake} U}{Q}$

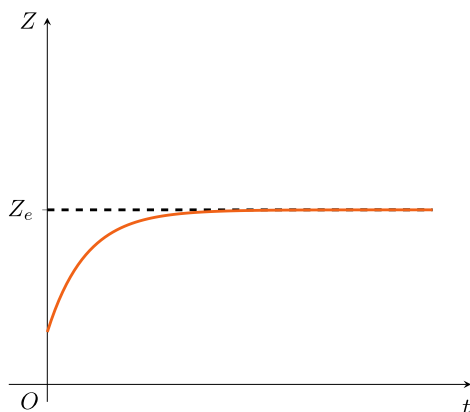
☒ $1 + \frac{U}{QC_{in}}$ ✓

Submit

You have used 1 of 2 attempts

i Answers are displayed within the problem

The amount of pollution in the lake, after the factory has started polluting has the following graph:



Now we reverse the situation. Suppose that the factory has been polluting the lake for a long time. Then one day, the factory finds a better way to dispose of their waste, and U is set to zero.

Exercise B

Question

What will the graph of Z look like after the factory stops polluting?



Answer



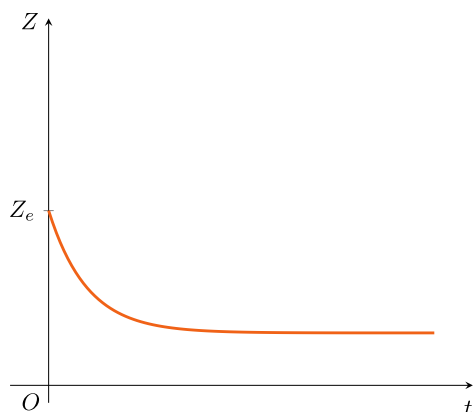
Reflection



Results

Correct Answer

This graph, because...



At the moment the pollution stops, Z will return to that in the initial situation, when just the river is polluting. The way Z decreases is symmetrical to the way it increased when the factory started polluting.

Your initial answer: This graph, because... (Option 3)

👤 "There will be a sharp drop to the initial value (without the factory's pollution)."

Your final answer: This graph, because... (Option 3)

👤 "There will be a sharp drop to the initial value (without the factory's pollution)."

See how your classmates answered below

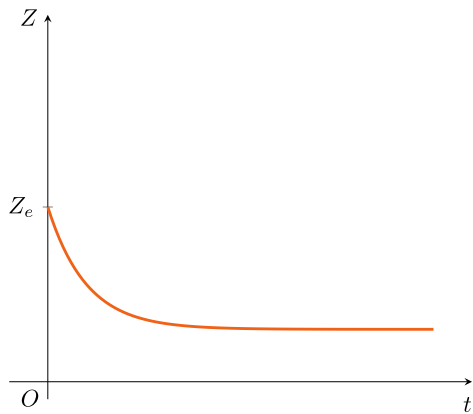
Class Breakdown

This is a look at how your classmates answered the question during the initial and final rounds.

Answer Options

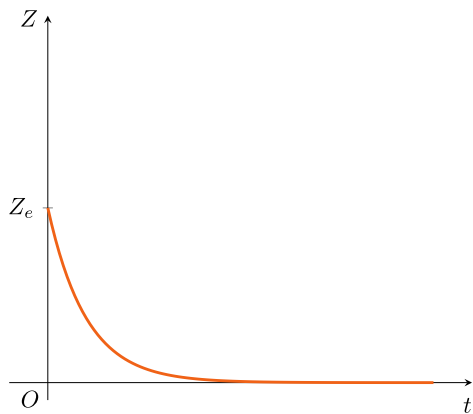
Option 1 (correct)

This graph, because...



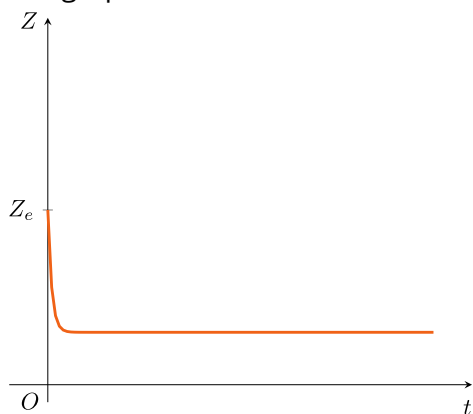
Option 2

This graph, because...



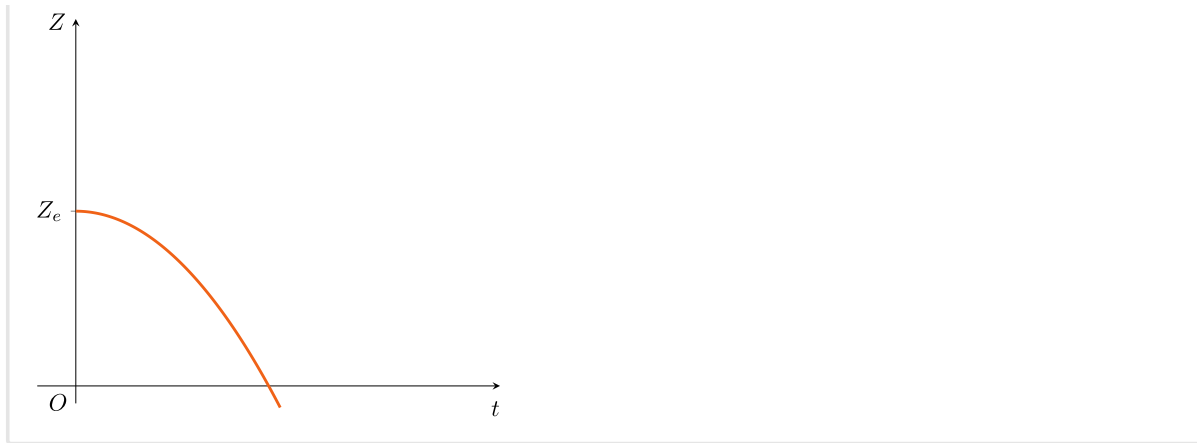
Option 3

This graph, because...



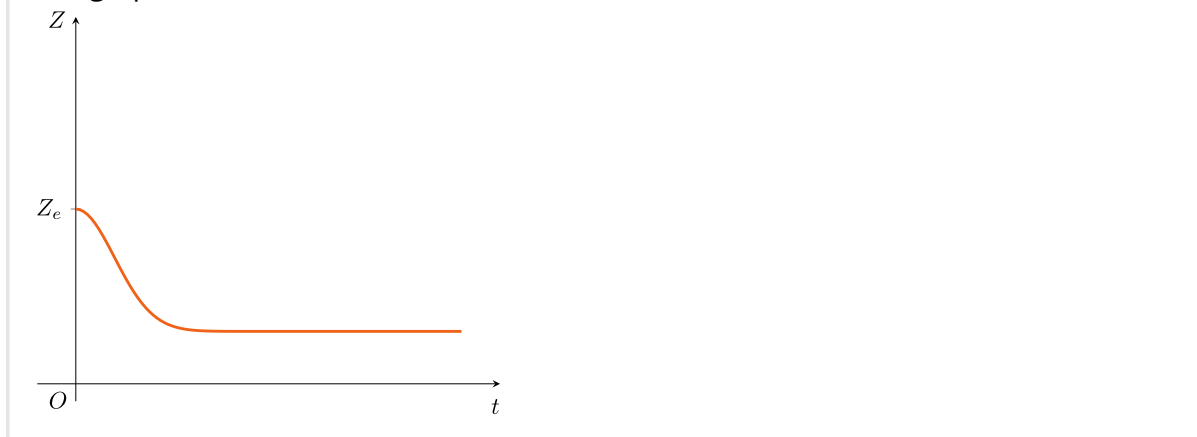
Option 4

This graph, because...

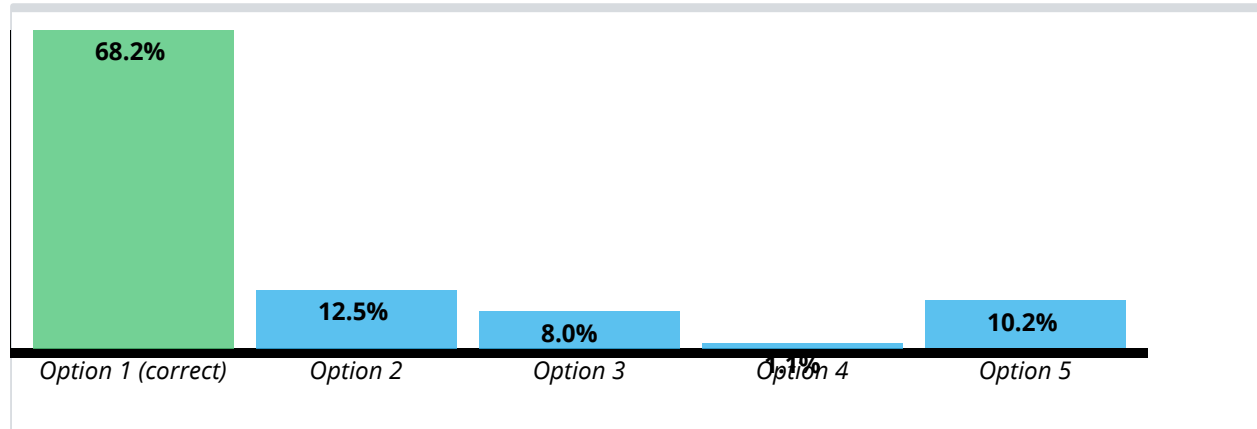


Option 5

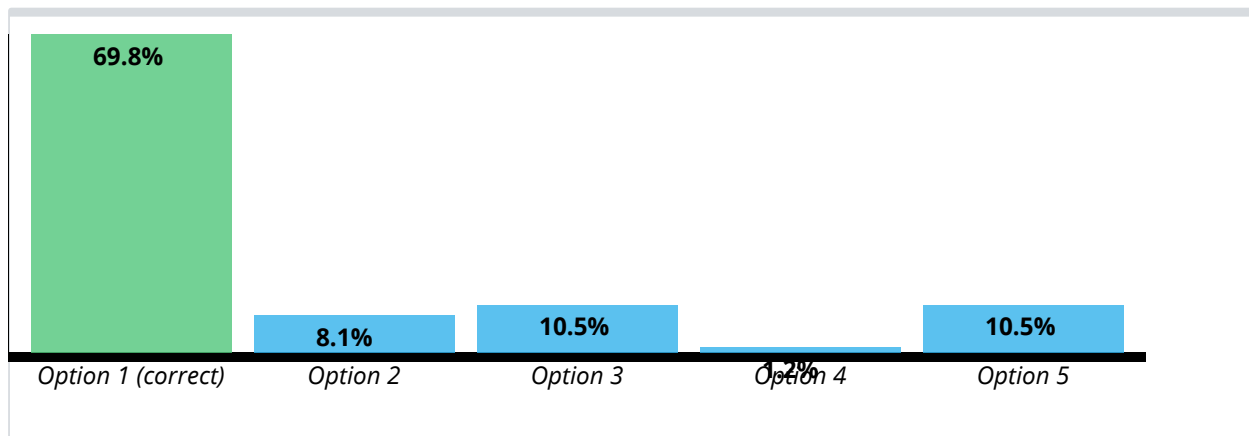
This graph, because...



Initial Answer Selection



Final Answer Selection



On the next page, we invite you to do some more critical thinking about this mixing problem.

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